

ASSIGNMENT-2

Q.1 Explain the following types of database security issues

1. Legal and ethical issues
2. Policy issue
3. System-related issues

- Ans
1. Legal and ethical issues: Concern laws and ethical use of data. Organizations must protect user privacy and follow data protection regulations
 2. Policy Issues: Security rules defined by an organization that specify who can access the database and what actions they can perform.
 3. System-related issues: security problems caused by system weakness such as software bugs, weak authentication, hacking or malware

Q.2 Explain the following database security control techniques: 1. Access Control 2. Inference control 3. Flow control 4. Data encryption

1. Access control: Restricts database access to authorized users using permissions
2. Inference control: Prevents users from deducing confidential information from accessible data
3. Flow control: Controls the flow of information to ensure data does not move to unauthorized users
4. Data encryption: converts data into coded form so only authorized users can read it

3. consider a table `Employee` (`EmpID`, `EmpName`, `Salary`, `DeptID`). Write an SQL query to create a view that displays only `EmpID`, `EmpName`, and `DeptID` from the `Employee` table.
- .view is a virtual table created from query

```
CREATE VIEW Employee-View AS → (creates a view)
SELECT EmpID, EmpName, DeptID → (columns to display)
FROM Employee ; → (source table)
```

<https://leetcode.com/problems/employee-bonus>

The screenshot shows the LeetCode problem page for '577. Employee Bonus'. The problem is categorized as 'Easy'. The description includes two tables: 'Employee' and 'Bonus'. The 'Employee' table has columns: empId (int), name (varchar), supervisor (int), and salary (int). The 'Bonus' table has columns: empId (int) and bonus (int). The problem asks to write a MySQL query to find the bonus of employees whose bonus is less than 1000 or is null. The code editor on the right shows the following SQL query:

```
1 # Write your MySQL query statement below
2 SELECT e.name, b.bonus
3 FROM Employee e
4 LEFT JOIN Bonus b
5 ON e.empId = b.empId
6 WHERE b.bonus < 1000 OR b.bonus IS NULL;
```

The test result section shows 'You must run your code first'.

<https://leetcode.com/problems/managers-with-at-least-5-direct-reports>

The screenshot shows the LeetCode problem page for '570. Managers with at Least 5 Direct Reports'. The problem is categorized as 'Medium'. The description includes the 'Employee' table with columns: id (int), name (varchar), department (varchar), and managerId (int). The problem asks to write a MySQL query to find managers with at least five direct reports. The code editor on the right shows the following SQL query:

```
1 # Write your MySQL query statement below
2 SELECT e1.name
3 FROM Employee e1
4 JOIN Employee e2
5 ON e1.id = e2.managerId
6 GROUP BY e1.id, e1.name
7 HAVING COUNT(e2.id) >= 5;
```

The test result section shows 'Accepted' with a runtime of 78 ms. The input data is as follows:

id	name	department	managerId
101	John	A	null
102	Dan	A	101
103	James	A	101
104	Amy	A	101
105	Anne	A	101
106	Ron	B	101